

scores try to account for the fact that translations may vary in word usage and syntax from the source sequences. In our task, the predicted recipes must be correct across the channels and functions. Additionally, the order of predicted sequence matters as each position in sequence corresponds to a specific recipe component. In addition to the recipe component accuracy scores (e.g. Trigger Channel accuracy or Event Function accuracy), we consider the overall sequence accuracy and the positional accuracy of the predicted sequences. The sequence accuracy measures if the overall sequence is correct. The prediction is assigned a score of 1 only if the predicted string exactly matches the reference string. We expect the sequence score to be lower in general as the predictions have a higher level of scrutiny.

The positional score provides partial credit to the prediction, if the tokens correctly match the reference tokens at their position in the recipe. For example, if a prediction correctly identifies 3 of the 4 recipes components, it will receive a score of .75. We provide the distribution of positional errors in Table 4.

6 Results

We were able to successfully train all three architectures and had varying levels of success. In Table 3 we see that none of the seq2seq models were able to match the reported channel/function accuracy values in ?? and ??. Both previous works only reported aggregate channel and function accuracy values. For simplicity, we assumed those aggregate values were the same across channels and respectively. Of the seq2seq models, OpenNMT consistently performed better than our baseline and the Transformer model. The seq2seq models did fairly well on Action Channel and Action function prediction, coming within 10 points of the ?? scores. We believe this may be related to how recipes are described in general. In both IFTTT and Zapier, users are more explicit in describing the action scenario (the "Then") part of the program than they are in describing the Triggering event.

In contrast to IFTTT, the seq2seq models all performed strongly on the Zapier dataset (Table 2). Our baseline LSTM encoder-decoder that had an overall sequence accuracy of 84.75% and strong performance on predicting each of the component recipes parts in context of the full recipe. The Transformer model performs the best is 9 points better than the baseline model in sequence accuracy. We saw significant improvement over Event Function (6 point increase) and Action Function (4 point increase) identification. Overall we found the quality of the Zapier titles and descriptions better than those of IFTTT. Zapier titles were consistent their ability to concisely describe the recipes and the descriptions were also well written and provided useful clues

7 Discussion

Overall the seq2seq models were able to learn If-Then recipes in their entirety. The models were surprisingly robust and able to capture various linguistic variations and ambiguities found in the natural language descriptions. Across both datasets, the models tended to have higher Action and Action Function scores than Trigger/Event scores. From super-

Errors	Zapier			IFTTT		
	LSTM	ONMT	Trans.	LSTM	ONMT	Trans.
Zero	84.75	93.23	9.39E-01	52.35	55.42	53.07
One	10.54	5.00	4.73E-02	14.44	12.09	13.36
Two	3.13	12.6	9.26E-03	23.10	21.84	22.92
Three	0.68	0.21	6.17E-04	2.35	3.97	2.71
Four	0.91	0.31	3.70E-03	7.76	6.68	7.94

Table 4: Distribution of errors across all predictions by domain.

ficial analysis, we found that users on average were more specific in describing the action then they were describing the triggering event.

We had significant difficulty working with the IFTTT dataset. Nearly 50% of the train and validation sets were thrown away, and nearly 90% of the test set. Through visual and ad-hoc analysis, we would often find mistakes in both the annotated test set recipes and the scraped recipes. Given the age and volatility of the recipe urls, we were not confident that our experiment conditions matched those from previous works. We were unable to reproduce any of the prior results and therefore had difficulty doing a more thorough error analysis on our findings.

In Table 4, we provide the distribution across all the predictions on the respective tests sets. It is interesting to note that on the IFTTT dataset, the model likely to make two errors as opposed to one, three or four. We believe there is a tight coupling between trigger channel and trigger function predictions. We hypothesize that if the model fails to predict the Trigger, it will also fail to predict the Trigger Functions.

The seq2seq model's performance on the Zapier dataset was very encouraging. As a next step we plan to investigate argument extraction on the dataset. Additionally, we are interested in explore the transfer learning potential given quality disparity between Zapier and IFTTT. Both have similar vocabularies and recipe domains. We hypothesize a model trained on Zapier data may perform better on the IFTTT dataset than the model trained solely on IFTTT data. Finally, we are interested in investigating more complex program synthesis challenges. We believe more complex program representations can potentially be learned through seq2seq models.

8 Conclusion

In this paper we proposed modeling If-Then program synthesis as sequence learning task. The models attempted to learn how translate natural language descriptions into IFTTT and Zapier recipes. Three seq2seq architectures were evaluated: a baseline LSTM encoder-decoder, the OpenNMT Stacked RNN, and the Transformer model. The models were successfully trained and able to predict the full automation recipes in an end-to-end manner. Due to several challenges with IFTTT dataset, we found the seq2seq model performance to be adequate but unable to match the accuracy scores of prior work. Of the seq2seq models, the OpenNMT model performed the best on the IFTTT dataset, with a sequence accuracy score of 55%, positional accuracy of 75 %, and overall Action Channel accuracy score of 79%. In con-

trast, the seq2seq models performed strongly on the Zapier dataset. The Transformer model score the highest across all